

SERIGNE FALLOU MBACKE

Junior Data Analyst BI & Data Visualization

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🌐 LinkedIn 🐙 GitHub 🌐 Portfolio

Professional Summary

Junior Data Analyst completing a college diploma in Applied Data Science, passionate about transforming raw data into actionable business insights. Proficient in the full analytics pipeline from data cleaning and modeling with Python/SQL to building interactive dashboards in Power BI. Detail-oriented and eager to apply data visualization and analytical skills in a dynamic professional environment.

Technical Skills

Visualization & BI	Power BI (DAX, Power Query, Star Schema), Excel (Pivot Tables, VLOOKUP), Matplotlib, Seaborn
Languages & Analysis	Python (Pandas, NumPy, Scikit-learn, Scipy), SQL (complex queries), R
Databases	MySQL, MongoDB
Machine Learning	Scikit-learn, Collaborative Filtering, TF-IDF, Linear Regression
Tools	Jupyter Notebook, Git, Jira (Agile/Scrum), REST API

Academic Projects

Customer Profitability Analysis (CLV) Power BI, DAX, Power Query

- Built a star schema data model in Power BI with 6 related tables (100 clients, 10,000 transactions, 27 products) simulating a real-world banking use case.
- Cleaned and transformed raw data using Power Query; computed RFM scores in DAX to segment customers into 6 behavioral profiles.
- Calculated Customer Lifetime Value (CLV) per client and per segment, revealing that 20% of customers generate 53% of total revenue.
- Designed a 4-page interactive report: Global CLV Overview, RFM Segmentation, Product Analysis, and Cross-Selling Opportunities.

Book Recommendation System Analysis & Visualization Python, Scikit-learn, Scipy

- Explored and visualized a dataset of 196,000 user-book interactions (898 books, 48,000 users): rating distribution, Lorenz curves, popularity bias detection, and sparsity matrix visualization (~99.6%).
- Implemented an Item-Based KNN collaborative filtering pipeline using a sparse matrix (`csr_matrix`) and cosine similarity.
- Content-based recommendation using TF-IDF on 96 books (title, genres, description), enhanced with semantic embeddings via Jina AI REST API.
- Produced advanced visualizations (log histograms, user segmentation, user-item matrix overview) to communicate insights to stakeholders.

Predictive Analysis of Real Estate Prices R, Statistics

- Exploratory analysis to identify correlations between variables (area, location) and sale prices.
- Built a multiple linear regression model achieving $R^2 > 0.85$ on test data.
- Statistical interpretation of results to isolate the most influential factors.

- Designed and built a relational database for a transportation company (10+ tables with integrity constraints).
- Wrote complex SQL queries (joins, CTEs, window functions) to extract strategic KPIs (occupancy rates, most profitable routes).

Work Experience

Fishmonger

IGA Gatineau, QC

May 2024 Present

- Daily inventory management and stock tracking in the seafood department.
- Strict compliance with food safety and quality standards in a regulated environment.
- Customer service in a bilingual setting (French/English).

Cashier

Hess' Your Independent Grocer Ottawa, ON

September 2023 May 2024

- Processed daily transactions and managed cash register reconciliation.
- High-volume customer service in a bilingual environment (French/English).

Education

Diploma of College Studies (DEC) Applied Data Science

Collège La Cité Ottawa, ON

2024 2026 (in progress)

Languages & Soft Skills

Languages	French (Native) English (Functional/Technical B2)
Soft Skills	Analytical Thinking, Problem Solving, Teamwork, Autonomy